

OPEN SOURCE LLM AND MCP ACTIONS ON YOUR OWN COMPUTER

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Development and experimentation



8GB image



1GB image



400MB
download



GBs & complex
dependencies

Production deployment



LlamaEdge

1MB to 50MB total size
No external dependencies



LlamaEdge

- Model agnostic
 - LLMs, embedding models, VLMs, whisper, piper, Stable Diffusion, Flux, GPT-SoVITS
- Runtime agnostic
 - CUDA, WebGPU, Metal, MLX etc.
- Hardware agnostic
 - CPU, Nvidia, AMD, Mac Ms etc.
- Focus on deployment
 - Lightweight (<50MB total)
 - Minimum dependencies
 - No root, no daemon
 - Single portable binary file
 - Supports a wide range of hardware

LlamaEdge provides a developer framework – you can build apps that combine your own LLM, knowledge, and actions.

TRY IT!

Install Gaia

```
curl -sSfL 'https://github.com/GaiaNet-AI/gaianet-node/releases/latest/download/install.sh' | bash
```

Init with Qwen-Coder-2.5-0.5B LLM

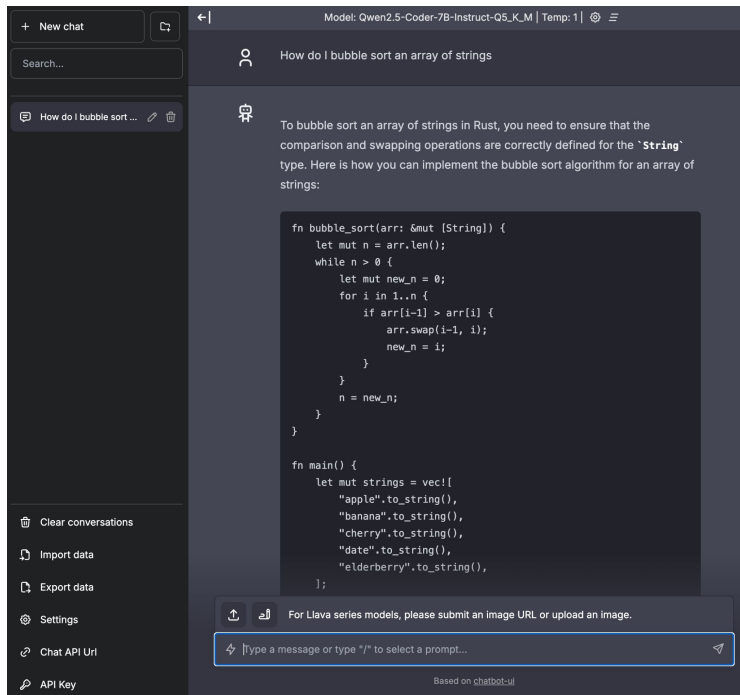
```
gaianet init --config https://raw.githubusercontent.com/GaiaNet-AI/node-configs/main/qwen-2.5-coder-0.5b-instruct/config.json
```

Start the local API server

```
gaianet start
```

Chatbot UI

<http://localhost:8080/chatbot-ui/index.html>



Make an API request

```
curl http://localhost:8080/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{
  "messages": [
    {
      "role": "system",
      "content": "You are a helpful assistant."
    },
    {
      "role": "user",
      "content": "Write a Rust function to sort an
array"
    }
  ]
}'
```

WHAT IS A GAIA NODE?

Eye and ears for the LLM – pre-query context

- A built-in vector database to ground the LLM with domain-specific facts
- Semantic search + internet search before answering each query

Example:

<https://huggingface.co/datasets/gaianet/learn-rust>

Try it:

https://llamaedge.github.io/chatbot-ui/index?api_url=https://rustcoder.gaia.domains

Hands and feet for the LLM – post query actions

- Ability to handle tool calls:
<https://docs.gaianet.ai/tutorial/tool-call/>
- Ready for MCP integration

WHAT IS THE GAIA NETWORK?

- A network of Gaia nodes that can offer services to the public
- Organized by domains
 - Domain are load balancers across identical nodes
 - It verifies all nodes are running the correct software
 - It charges users and distributes revenue to nodes
- Example
 - <https://rustcoder.gaia.domains>
 - <https://llama70b.gaia.domains>
- Start your own node and join the network!
 - <https://docs.gaianet.ai/getting-started/quick-start>
 - <https://docs.gaianet.ai/getting-started/register>

MCP USE CASES

PR REVIEW AGENT – USE MCP TO CONNECT TO GITHUB

GitHub PR Reviewer

LLM Configuration: Gaia — Qwen Coder 32b

LLM API Key: Optional

* You can get our API key through [this documentation](#).

GITHUB PR URL

Instructions on what and how to conduct the review

☒ Use llms.txt in the repo

Submit

<https://code.flows.network/webhook/ldDI5jrutZEAguuMPsS5>

INTRODUCING RUST CODER (WIP)

USE MCP TO CONNECT TO RUST COMPILER

[HTTPS://GITHUB.COM/WASMEDGE/WASMEDGE/ISSUES/4038](https://github.com/wasmEdge/wasmEdge/issues/4038)

🔥 Usage

The API provides the following endpoints:

🚀 Generate a Project

Endpoint: POST /generate

📄 Request Body:

```
{
  "description": "A command-line calculator in Rust",
  "requirements": "Should support addition, subtraction, multiplication, and division"
}
```

📄 Response:

```
{
  "project_id": "123e4567-e89b-12d3-a456-426614174000",
  "status": "processing",
  "message": "Project generation started"
}
```

🔍 Check Project Status

Endpoint: GET /project/{project_id}

📄 Response:

```
{
  "project_id": "123e4567-e89b-12d3-a456-426614174000",
  "status": "completed",
  "message": "Project generated successfully",
  "files": ["Cargo.toml", "src/main.rs", "README.md"],
  "build_output": "...",
  "run_output": "..."
}
```

📁 Project Structure

```
Project3/
├── app/                # Application code
├── data/               # Data storage
├── docker-compose.yml  # Docker Compose configuration
├── Dockerfile          # Docker configuration
├── output/             # Generated project output
├── parse_and_save_qna.py # Q&A parsing utility
├── qdrant_data/        # Vector database storage
├── requirements.txt     # Python dependencies
├── templates/         # API templates
└── tests/
```

Our AI Tools



Rust Learning Assistant

An intelligent assistant specifically designed to help you learn Rust programming. Get answers to your questions, code explanations, and personalized learning paths.

[Learn More](#)

Rust Code Generator

Generate production-ready Rust code from natural language descriptions. Save time and focus on solving complex problems.

[Learn More](#)

Python to Rust Translator

Easily convert your Python code to Rust. Replace slow Python implementations with high-performance Rust while maintaining functionality.

[Learn More](#)

PR Reviewer

Expert code review for your Rust pull requests with human oversight. Designed for team leaders to maintain code quality with the AI model of your choice.

[Learn More](#)

IDE Integration

Comprehensive guides on integrating our tools with popular IDEs like Zed, Cursor, VS Code, and more. Coming soon: MCP Server for enhanced capabilities.

[Learn More](#)

Explain this PR to me

Translate complex Rust pull requests into clear, business-focused explanations for product managers and non-technical stakeholders.

[Learn More](#)

<https://lowcoderust.com/>

<https://openmcp.app/>

Features

Transform local MCP servers into secure, reliable, and profitable online services



Execution sandbox

The WasmEdge sandbox is an isolated and secure execution environment for MCP services. For example, it could limit access to certain local files and URLs.



Aggregate MCP servers

The OpenMCP proxy can dispatch requests to multiple local MCP servers, providing a single SSE URL for all functions in those MCP servers.



Security Rules

Implement and enforce security rules defining which commands and API calls are allowed or prohibited by each MCP server.



Monetization

Decentralized payment protocol supporting subscriptions and usage-based billing based on API keys.



Authentication

Define and implement access control for each MCP server behind the OpenMCP proxy.



Verifiable Services

Crypto staking for OpenMCP service providers to ensure that they are running the correct MCP functions.



THANK YOU